

Drone Object Detection Using YOLOv10

Advance Machine Learning – Final Project Report

Prepared By

Mohammad Ismaeel
Ahmad Nasasra

Instructor

Dr. Adnan Salman

Program

CAP-AI (Computer Science – Artificial Intelligence Track)

Department of Computer Engineering

An-Najah National University (NNU)

2026

1. Summary

This project presents a drone-based object detection system using YOLOv10 for detecting persons and cars in aerial images. The VisDrone2019 dataset was selected because it represents realistic drone-captured scenes with significant challenges such as small objects, crowded environments, scale variation, and occlusion. Transfer learning was applied by fine-tuning a pre-trained YOLOv10b model on a customized version of the dataset containing only two classes: person and car. Multiple experiments were conducted to investigate the impact of image resolution on detection performance. Three input resolutions were evaluated: 640×640, 960×960, and 1280×1280. The results demonstrate that increasing image resolution significantly improves object detection accuracy, particularly for small objects in aerial imagery. The best-performing model achieved a Precision of 0.806, Recall of 0.743, mAP@50 of 0.806, and mAP@50-95 of 0.519 using an input resolution of 1280×1280.

2. Introduction

2.1 Motivation

Object detection has become one of the most important applications of computer vision and deep learning. In recent years, drones have been increasingly used in surveillance, traffic monitoring, disaster management, smart city applications, and autonomous systems. These applications require reliable object detection systems capable of identifying vehicles and pedestrians from aerial imagery.

Detecting objects from drone images presents unique challenges compared to traditional ground-level images. Objects often appear very small, scenes can be highly crowded, and different viewing angles introduce additional complexity. Therefore, selecting an effective object detection model and understanding the factors affecting its performance are essential tasks.

This project investigates the use of YOLOv10 for aerial object detection and studies how image resolution influences detection accuracy.

2.2 Background

Object detection is a computer vision task that combines object classification and object localization. Unlike image classification, object detection must identify both the category and the position of objects within an image.

Deep learning has significantly advanced object detection through convolutional neural networks (CNNs). Among modern detectors, the YOLO (You Only Look Once) family has become one of the most widely used approaches due to its high speed and competitive accuracy.

YOLOv10 is a single-stage detector that predicts object locations and classes in a single forward pass. This design enables real-time performance while maintaining strong detection capabilities.

Another important concept used in this project is transfer learning. Instead of training a model from scratch, transfer learning starts from a model already trained on a large dataset. The pre-trained model is then fine-tuned on a specific task, reducing training time and improving convergence.

3. Dataset

3.1 Source

The dataset used in this project is the VisDrone2019 Object Detection Dataset. The dataset contains aerial images collected using drones in various urban environments under different weather, illumination, and scene conditions.

The dataset is publicly available and widely used for evaluating aerial object detection algorithms.

3.2 Dataset Description

The original VisDrone dataset contains multiple object categories including:

- Pedestrian
- Person
- Car
- Van
- Bus
- Truck
- Bicycle
- Motor
- Awning-tricycle
- Tricycle

For this project, only two categories were selected:

- Person
- Car

The dataset includes:

- High-resolution aerial images
- Scale variation
- Crowded scenes
- Partial occlusion
- Different viewing angles
- Real-world drone imagery

These characteristics make the dataset challenging and suitable for evaluating modern object detection systems.

3.3 Dataset Preprocessing

Several preprocessing steps were performed before training:

1. Conversion of VisDrone annotations into YOLO format.
2. Selection of Person and Car classes only.
3. Removal of all remaining classes.
4. Removal of ignored regions.
5. Conversion of bounding box coordinates into normalized YOLO coordinates.
6. Creation of training, validation, and testing datasets.

The classes were re-labeled as follows:

Original Class	New Label
Person	0
Car	1

This preprocessing simplified the detection task and allowed the model to focus on the two target classes.

4. Methodology

4.1 Overall Approach

The project uses YOLOv10b as the object detection framework.

The workflow consists of the following stages:

1. Dataset preparation.
2. Annotation conversion.
3. Class filtering.
4. Model training.
5. Validation.
6. Inference.
7. Performance evaluation.

Transfer learning was employed using pre-trained YOLOv10 weights. The model was then fine-tuned on the customized VisDrone dataset.

4.2 Training Configuration

The final training configuration was:

Parameter	Value
Model	YOLOv10b
Dataset	VisDrone2019
Classes	Person, Car
Epochs	100
Batch Size	12
Optimizer	AdamW
Image Resolution	1280×1280

Parameter	Value
Transfer Learning	Yes
Early Stopping	Enabled
Best Epoch	67

Several data augmentation techniques were applied during training, including:

- Mosaic augmentation
- Copy-Paste augmentation
- Random scaling
- Horizontal flipping
- Color augmentation

4.3 Resolution Study

One of the main objectives of this project was to study the effect of image resolution on detection performance.

Three experiments were conducted:

- 640×640
- 960×960
- 1280×1280

This comparison provides insight into the relationship between image resolution and object detection accuracy in aerial imagery.

4.4 Evaluation Criteria

The model was evaluated using standard object detection metrics:

Precision

Measures the proportion of correct detections among all predicted detections.

Recall

Measures the proportion of ground-truth objects successfully detected.

mAP@50

Mean Average Precision at an IoU threshold of 0.50.

mAP@50-95

Mean Average Precision averaged over multiple IoU thresholds ranging from 0.50 to 0.95.

These metrics are commonly used to evaluate object detection performance and provide a comprehensive assessment of both classification and localization quality.

5. Results

5.1 Resolution Comparison

To analyze the effect of image resolution, three experiments were conducted.

Resolution	Precision	Recall	mAP@50	mAP@50-95
640×640	0.689	0.521	0.544	0.310
960×960	0.742	0.601	0.652	0.396
1280×1280	0.806	0.743	0.806	0.519

The results clearly show that increasing image resolution improves overall performance.

5.2 Final Model Performance

The best-performing model was obtained using a resolution of 1280×1280.

Metric	Value
Precision	0.806
Recall	0.743
mAP@50	0.806
mAP@50-95	0.519

Class-wise Results

Class	mAP@50
Person	0.713
Car	0.900

5.3 Visual Results

The following figures should be included in this section:

- Training Results Curve (results.png)
- Precision-Recall Curve
- F1 Curve
- Confusion Matrix
- Normalized Confusion Matrix
- Sample Predictions

These visualizations provide additional insight into model behavior and performance.

5.4 Discussion

The results indicate that image resolution is a critical factor in aerial object detection.

The model trained using 1280×1280 images significantly outperformed the models trained using lower resolutions. Higher resolutions preserve more spatial information, making it easier to detect small objects.

Vehicle detection achieved excellent performance with an mAP@50 of 0.900. Cars occupy larger regions in drone images and exhibit more consistent visual features.

Person detection achieved an mAP@50 of 0.713. Although lower than car detection, this result is reasonable given the challenges associated with detecting small pedestrians in crowded aerial scenes.

The confusion matrix demonstrates strong classification capability, while the training curves indicate stable convergence without significant overfitting.

Overall, the experiments confirm that high-resolution inputs substantially improve detection performance in drone imagery.

6. Conclusion

This project successfully developed a drone-based object detection system using YOLOv10 and the VisDrone2019 dataset. Transfer learning enabled efficient training and strong detection performance.

The study investigated the impact of image resolution on object detection accuracy. Experimental results showed that increasing the image resolution from 640×640 to 1280×1280 significantly improved Precision, Recall, and mAP scores.

The final model achieved a Precision of 0.806, Recall of 0.743, mAP@50 of 0.806, and mAP@50-95 of 0.519. Vehicle detection achieved excellent accuracy, while person detection remained more challenging due to the small-object nature of aerial imagery.

The project demonstrates the effectiveness of YOLOv10 for drone-based object detection and highlights the importance of image resolution when dealing with small objects.

7. Future Work

Future improvements may include:

- Training on larger aerial datasets.
 - Evaluating additional YOLO variants.
 - Comparing YOLOv10 with transformer-based detectors.
 - Applying advanced small-object detection techniques.
 - Deploying lightweight models on embedded drone hardware.
 - Investigating real-time edge inference systems.
-

8. References

- [1] VisDrone Dataset. <https://github.com/VisDrone/VisDrone-Dataset>
- [2] Ultralytics Documentation. <https://docs.ultralytics.com>
- [3] PyTorch Documentation. <https://pytorch.org>
- [4] Wang, A., et al. YOLOv10: Real-Time End-to-End Object Detection.
- [5] Redmon, J., et al. You Only Look Once: Unified, Real-Time Object Detection.